

Applying Montessori's and Piaget's theories of child development to their effect on curriculum building in early years education

The main claim for the relevance of Piaget's four developmental stages of the child and Montessori's theory of child development (Aubrey and Riley, 2022) to the structure and formation of the curriculum in early years education is that they both contribute to the building of curricula in various ways. However, it can be argued that Montessori's theory is more relevant and progressive in the UK for teaching early years, having a “*meaningful and positive impact on child outcomes*” (Randolph et al. 2023, p.2) and relating more to the needs of the child and learning outcomes today more mainstream education for example, which reflects Piaget's learning theory more.

Montessori developed her theory of the stages of child development through careful observation of how children's bodies and minds change at different age stages (Ross, 2012). These are known as planes of development (Aubrey and Riley, 2022). These are the absorbent mind, involving exploration of the world through the senses (ages 0-6), the conscious mind involving reasoning (ages 6-12), abstract thinking where learning is about a person's position and contribution to the society they live in (ages 12-18), and finally early adulthood (ages 18-24) where a person enhances their specialist skills and becomes fully independent (Grazzini 2004). When focusing on the first plane of development, Montessori's ideas are crucial in understanding how the school curriculum should be shaped so that very young children can build a consistent skill set that can become increasingly more complex over time (Grazzini, 2004). The Montessori

education system uses these stages to shape the levels of complexity in the curriculum and is characterised by allowing children to have choice and independence in learning. Their play should also mirror actions and events they could perform in everyday life. A Montessori classroom will have a teacher that encourages and helps students to learn by creating an environment that assists them during the different planes of development , such as providing more sensory learning materials in the absorbent stage (Isaacs, 2018).

Moreover, Chattin-McNichols (1981) studied the effect of Montessori education on preschool children, finding that “*all the children in their first year of preschool showed a statistically significant gain in Stanford-Binet IQ*” (Chattin-McNichols, 1981, p.4) which is a test taken by children to determine their level of understanding about different subject areas. This displays the positive effect of Montessori education on young children. However, Chattin-McNichols (1981) also discovered that overall, the preschoolers didn't increase their IQs in the following two years using the same Stanford-Binet test, perhaps due to the lessening effect of the Montessori curriculum overtime. It can still be expressed that the Montessori curriculum is highly effective in promoting learning at the beginning of preschool and that children understand subjects such as vocabulary learning better than the control group students who didn't receive Montessori education (Miller and Dyer, 1975 cited in Chattin-McNichols, 1981).

In addition, Montessori believed that in the past young children had been “*working from a curriculum designed for societal interest rather than the needs of the child*” (Aubrey and Riley, 2022, p.28). Therefore, she emphasises the importance of developing a child-centered curriculum to meet the educational needs of children in accordance with the progression of their

developmental stages (Yadav, 2023). Learning through play is a vital part of Montessori education as it is “*child centered, constructivist, affectively positive, and hands-on*” (Lillard, 2013, p.2) and differs from the focus of the national curriculum of the UK by developing and emphasising its “*own individual pedagogic bases*” (Kirkham and Kidd, 2017 p.21) of creative freedom. However, Montessori's theory of learning deemphasised the importance of imaginative play, concentrating more on the idea of lessons being interactive between young children and teachers, and on children having a certain level of choice about what to play with in the classroom (Lillard, 2013). We can see that at the early stages of child development, such as in the absorbent stage, it is necessary for children to be able to explore their senses which will help their cognitive expansion and allow them to gain greater independence (Hiltunen 2017).

Montessori advocated for sensory learning materials being used in early years education, such as sandpaper tracing in alphabet learning, to understand the formation of letters through the sense of touch and ultimately help with letter recognition. Norman (2020) found that 76% of the children they studied who used the sandpaper letters technique of learning increased their letter sound identification significantly. This demonstrates that the Montessori approach to learning using sensory objects during the absorbent stage can be beneficial in children's development.

Furthermore, Piaget also describes four different stages of child development, with more focus on the child's cognitive development. Piaget developed his ideas by theorising that a child's mind develops certain operations, or rules about the world, through the cognitive maturation of the brain and not necessarily due to the events they experience (Aubrey and Riley, 2022). Similarly to Montessori's theory, Piaget's four stages are chronological by age as this reflects the ages in which the brain takes a new developmental step. These stages are the sensorimotor stage (ages 0-

2) where the process of developing sensory awareness starts, the preoperational stage (ages 2-7) where a child can think symbolically through imaginative thought and play but does not yet have the ability to think abstractly, the concrete operational stage (ages 7-11) where a child can have a more complex understanding of abstract ideas and can take into consideration other people's perspectives, and lastly the formal operational stage (ages 12-18+) where the child now has a full understanding of abstract thinking and can solve problems outside of concrete issues (Soran University et al., 2019). Schools that apply Piaget's child development progression into their curriculum will gradually increase the difficulty and complexity of the ideas they present students. Strauss (1972) noted that as a child reaches the formal operational stage from 12 years and beyond, they are now able to engage with more complex mathematical problems for example as they have the mental capabilities and sufficient capacity to understand abstract thought. In the UK students begin secondary school around this age, displaying that they are now able to transition into understanding more complex concepts, which follows Piaget's development stages closely.

Additionally, Piaget theorised that children construct a path of self-discovery and progress in their cognitive development by experimenting and playing with real and concrete objects during the preoperational stage, which then allows them to visualise abstract ideas in their minds and begin to engage in make believe play (Bashrin, 2015). As a child grows out of the sensorimotor stage of infancy which is primarily focused on the physical needs that allow them to survive, such as eating and sleeping, they start to develop a curiosity about toys around them as well as learning experiences presented to them by adults in educational settings (Halpenny and Pettersen, 2013). This self-directed play is important in Piaget's constructivist methodology and

in the stage of preoperational cognitive development in early years educational curriculums. The growth to more abstract and conceptual thinking as the child progresses through the stages is outlined by Kamii and Radin (1967), who state that in an early years teaching curriculum it is important to allow a transition from concrete object representation to being able to assess objects and ideas that the child can't see but can think about. Kamii and Radin (1967) describe how a child can be helped to develop an understanding of what part of an object, such as a picture of a cows head, means in relation to the rest of the body which is not visible, and can come to know that this object is a cow regardless of if the rest of its body is out of view. It is further emphasised that in order for the child to fully understand abstract thought they must be "*thoroughly acquainted with objects in a sensory-motor way*" (Kamii and Radin, 1967, p.317) and so should be encouraged to use and play with what they see around them independently, such as learning to become proficient in self-feeding with cutlery and learning to drink from a cup. The act of touching and feeling these objects and associating them with mealtimes means a child will develop a bank of mental images and representations in their minds as they build each new experience and repeat them over time. A child can then pretend they are eating and drinking when playing (abstract thinking) as they are now aware of how to do this, having practiced with concrete objects. Therefore, it can be said that unrestricted play and allowance of free exploration seems crucial for a child's development.

Moreover, Montessori's theory as applied to the curriculum is praised for its effect on the creativity of children, with some researchers discovering that students who were taught with the Montessori model "*produced stories that were significantly more creative than non-Montessori students' stories.*" (Fleming et al., 2019, p.3). Contradictorily, Aubrey and Riley (2022)

illustrated a critique of Montessori by John Dewey, who disagreed with her theory as he believed it actually impedes the creativity and imagination of children by creating a structured approach to the exploration of materials in the learning environment, with Montessori stating imaginative play is not very important in the education of a child (Roopnarine and Johnson, 2005 cited in Aubrey and Riley, 2022). Instead, Montessori believed that the early years curriculum should give children ‘Practical Life exercises’ where they could practice and act out scenarios they have seen at home like cooking, cleaning and building. Although there are toys present in a Montessori setting, they are made to be used in a practical way for children to learn how to use them in the real world (Lillard, 2013). Also, Montessori has been criticised in terms of the supposed lack of discipline her theory creates in that her focus on the autonomy of the child may lead to an unstructured curriculum and bad behaviour in the classroom (Yadav, 2023). Also, it has been noted that there is a contradiction within Montessori's theory in that she discusses specific developmental stages but then advocates for mixed age classrooms, which may negatively affect children who are “*developmentally ahead or behind their peers.*” (Yadav, 2023, p.32).

Furthermore, Piaget has also been critiqued for his cognitive development theory as applied to the early year's curriculum. Montessori critiqued Piaget's theory of the stages of child development as she believed, unlike Piaget, that there was no correct age that a child needed to pass through each stage but rather that these development stages were “*sensitive periods*” (Aubrey and Riley, 2022, p.35) and that each child would develop at different rates and express how they learn uniquely. In addition, Babakr et al. (2019) critique Piaget's notion of what a child can achieve in the preoperational stage when they are in the early years stage of the school

curriculum. Piaget noted that at this stage a child would not be able to conserve information about the amount of something, for example water in a vial, if the shape it is contained in changes, such as the water being poured into a different shaped vial (Franzoi, 2011 cited in Babakr et al., 2019). Gelman (1972, cited in Babakr et al., 2019) describes a study conducted in which the difficulty of the conservational tasks of changing the shape of an object was increased, and children as young as 3 years old could undertake the tasks well and had the ability to understand complex ideas around why the amounts stayed the same while the objects changed. This critique of Piaget could affect the curriculum as Piaget's concept may mean teachers are underestimating what young students can achieve, and the curriculum may therefore not be suitable to their abilities. This is why Montessori's use of an open classroom with plenty of choice for children to decide what to interact with, rather than the restrictions seen in a Piagetian curriculum, may be better. Montessori's method allows for children to develop at their own rate and use objects around them in a way that encourages independence through play.

In summary, both Piaget's and Montessori's approaches to the development of the early year's curriculum are valuable and can each help educators to understand how best to facilitate learning. However, through the evidence presented, the Montessori approach to child development seems to be most beneficial and most accessible for teachers and students. While the theory does lack focus on the imaginative parts of play in learning, it produces excellent outcomes in children's independence and ability to think more abstractly through sensory and realistic resources in the classroom. Additionally, while Piaget's theory of cognitive development is highly insightful in its advocacy of self-discovery and the development of abstract thinking in early years education, it seems too rigid and deterministic in its description

of the different stages of development, with Montessori's sensitive periods better capturing the reality of how each child progresses at their own pace when learning.

References

Aubrey, Karl, and Alison Riley. *Understanding and Using Educational Theories*. 3rd ed, SAGE Publications, 2022.

Babakr, Zana Hasan, et al. 'Piaget's Cognitive Developmental Theory: Critical Review'. *Education Quarterly Reviews*, vol. 2, no. 3, Sept. 2019. *DOI.org (Crossref)*, <https://doi.org/10.31014/aior.1993.02.03.84>.

Bashrin, Syeda Dishari. *Piaget's Pre-Operational Stage in Children: A Comparative Study*. 2015.

Chattin-McNichols, John P. 'The Effects of Montessori School Experience'. *Young Children*, vol. 36, no. 5, 1981, pp. 49–66. *JSTOR*, <https://www.jstor.org/stable/42642922>.

Fleming, David J., et al. *The Montessori Model and Creativity*. 2019.

Grazzini, Camillo. 'The Four Planes of Development '. *NAMTA Journal* , 2004.

Halpenny, Ann Marie, and Jan Pettersen. *Introducing Piaget: A Guide for Practitioners and Students in Early Years Education*. Routledge, 2013, <https://doi.org/10.4324/9780203111895>.

Hiltunen, Hannele. *Play and Motivation : Through the Eyes of Montessori, Vygotsky, Deci and Ryan*. 2017.

Isaacs, Barbara. *Understanding the Montessori Approach: Early Years Education in Practice*. Second edition, Routledge, Taylor & Francis Group, 2018.

Kamii, Constance K., and Norma L. Radin. 'A Framework for a Preschool Curriculum Based on Some Piagetian Concepts ¹'. *The Journal of Creative Behavior*, vol. 1, no. 3, July 1967, pp. 314–24. *DOI.org (Crossref)*, <https://doi.org/10.1002/j.2162-6057.1967.tb00032.x>.

Kirkham, Julie Ann, and Evan Kidd. ‘The Effect of Steiner, Montessori, and National Curriculum Education Upon Children’s Pretence and Creativity’. *The Journal of Creative Behavior*, vol. 51, no. 1, Mar. 2017, pp. 20–34. *DOI.org (Crossref)*, <https://doi.org/10.1002/jocb.83>.

Lillard, Angeline S. ‘Playful Learning and Montessori Education’. *NAMTA Journal*, vol. 38, no. 2, 2013, pp. 137–74. *ERIC*, <https://eric.ed.gov/?id=EJ1077161>.

Lillard, Angeline S., and Jessica Taggart. ‘Pretend Play and Fantasy: What If Montessori Was Right?’ *Child Development Perspectives*, vol. 13, no. 2, June 2019, pp. 85–90. *DOI.org (Crossref)*, <https://doi.org/10.1111/cdep.12314>.

Norman , Elizabeth. *Cultivating Toddler Emergent Literacy Behaviors Using the Montessori Sandpaper Letters*. 2020, <https://sophia.stkate.edu/cgi/viewcontent.cgi?article=1359&context=maed>.

Randolph, Justus J., et al. ‘Montessori Education’s Impact on Academic and Nonacademic Outcomes: A Systematic Review’. *Campbell Systematic Reviews*, vol. 19, no. 3, Aug. 2023, p. e1330. *PubMed Central*, <https://doi.org/10.1002/cl2.1330>.

Ross, Suzanne. ‘The Montessori Method: The Development of a Healthy Pattern of Desire in Early Childhood’. *Contagion: Journal of Violence, Mimesis, and Culture*, vol. 19, 2012, pp. 87–122. *JSTOR*, <https://www.jstor.org/stable/41925335>.

Soran University, et al. 'Piaget's Cognitive Developmental Theory: Critical Review'. *Education Quarterly Reviews*, vol. 2, no. 3, Sept. 2019. DOI.org (Crossref), <https://doi.org/10.31014/aior.1993.02.03.84>.

Strauss, Sidney. 'Learning Theories of Gagne Piaget: Implications for Curriculum Development'. *Teachers College Record: The Voice of Scholarship in Education*, vol. 74, no. 1, Oct. 1972, pp. 1–17. DOI.org (Crossref), <https://doi.org/10.1177/016146817207400102>.

Yadav, Dr Neeraj. 'Critique of Montessori's Method of Early Childhood'. *International Journal of Multidisciplinary Research in Arts, Science and Technology*, vol. 1, no. 5, Dec. 2023, pp. 29–36. [ijmrast.com](https://doi.org/10.61778/ijmrast.v1i5.28), <https://doi.org/10.61778/ijmrast.v1i5.28>.